EE3001: Solid State Devices
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Assignment #2

Try yourself to solve the problems with detailed steps. Take a limited time quiz to test your understanding on related concepts.

Problems will be discussed during the tutorial class. You need to contact TAs or teacher if you are unable to do anything useful by yourself or with the help of your friends.

If needed, assume the following values by default unless otherwise specified: Si is the default semiconductor; SiO_2 is the default oxide; T @room temperature is 300 K; n_i@room temp. = $10^{10} cm^{-3}$; k_BT=0.026 eV (at room temperature); $E_{G,Si}$ = 1.12 eV and temperature independent; at 300 K, N_c=$2.8 \times 10^{19} cm^{-3}$ and N_v=$1.02 \times 10^{19} cm^{-3}$

1. (a) If N_c represents the density of states in a strip αkT wide near the edge of the conduction band, determine the value of α.

   (b) Obtain $(E_c - E_F)$ at room temperature for electron concentrations of (i) 5×10^{14} cm^{-3}, (ii) 5×10^{17} cm^{-3} and (iii) 10^{19} cm^{-3} in Si. Show the positions of E_F in the energy bandgap. Comment on the applicability of Boltzmann approximation in various cases.

2. (a) For a non-degenerate Si sample at room temperature, E_c - E_f = $4.1 k_BT$. Calculate the probability of finding an electron at an energy level of k_BT above the conduction band.

   (b) Find the doping concentration in the Si sample at T=300 K if all the dopant atoms are ionized.

3. In a semi-log axis the plot of $1/k_BT$ (linear x-axis in per eV) versus $n_i/\sqrt{N_cN_v}$ (log_e y-axis) for a semiconductor sample appears to be linear with a slope of -0.71 near T=300 K. What is the band-gap of the semiconductor?

4. The average kinetic energy of electrons in the conduction band is given by the following relation

$$< K.E >= \frac{\int_{E_c}^{\infty}(E - E_c)D(E)f(E)dE}{\int_{E_c}^{\infty} D(E)f(E)dE}$$

   For a non-degenerate semiconductor in which $f(E)$ can be approximated by the Maxwell-Boltzmann distribution, if the average kinetic energy is given by $\lambda\ k_BT$, find the value of λ. Here D(E) denotes the density of states in conduction band.

5. (a) For a pure Si sample, if the temperature is increased from 300 K to 400 K, find the ratio of the intrinsic carrier concentration, $n_i(400K)/n_i(300K)$.

(b) At 400 K, what are the values of n_i for (i) Ge (E_g = 0.66 eV, N_c = 10^{19} cm^{-3}, N_v = 6×10^{18} cm^{-3}) and (ii) GaAs (E_g = 1.42 eV, N_c = 4.7×10^{17} $cm-3$, $N_v = 7 \times 10^{18}$ cm^{-3})? Note that given values of N_c and N_v are in 300 K.

6. (a) A Si sample already doped with acceptor atoms ($N_A = 10^{14}$ cm^{-3}) is further doped with donor atoms (N_D cm^{-3}). Calculate the free carrier concentration (n and p) for (i) $N_D = N_A$; (ii) $N_D = 5N_A$; (iii) $N_A = 3N_D$;

   (b) Check and comment if the calculations of n and p are simplified when $|N_D - N_A| \ll 2n_i$ or $|N_D - N_A| \gg 2n_i$.

   (c) Plot the concentration of n and p as a function of N_D (varying from 10^{13} cm^{-3} to 10^{17} cm^{-3}) if N_A is fixed at 10^{15} cm^{-3}.

   (d) If the Si sample in (a) has no donor doping, calculate the ratio, p/n at (i) 300 K, (ii) 400 K, (iii) 500 K and (iv) 600 K and plot corresponding $E_F - E_i$ as a function of temperature.